

Odds-Inflation Anonymity: Posterior Odds Budgets, Composition, and Operational Limits

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Abstract

We formalize anonymity as a bound on *posterior odds inflation*: after observing a transcript, how many bits can the odds in favor of any single identity increase relative to a prior? This pointwise notion yields referee-checkable, per-output guarantees that (i) are closed under post-processing, (ii) compose sequentially under interaction, and (iii) imply concrete limits on identification, shortlisting, and property inference. We relate odds-inflation anonymity to pointwise maximal leakage and maximal leakage, and give practical conversions from common “equalization” contracts (e.g. total-variation closeness to a background distribution and Rényi-moment bounds). The result is a compact endpoint metric that cleanly consumes mechanism-specific toolboxes (e.g. metadata-hiding overlay compilers) while supporting modular certification.

Scope. We define a compact, mechanism-agnostic endpoint for anonymity: a bound on realized posterior odds inflation for every observable transcript. The goal is a referee-checkable per-output statement that composes under interaction and can be consumed by overlay-specific “compiler” analyses.

Units. Throughout, information budgets are measured in bits: we write $\log_2$ and 2^ϵ for realized odds-inflation / pointwise-maximal-leakage caps. Imported results stated in nats can be converted by dividing by $\ln 2$.

Imports. Mechanism-specific contracts (scheduling/compilers, random-walk delegation, and stop-time padding) are imported from the published backbone notes and cited only as black-box prerequisites. In particular, we treat termination-time hiding and TV certificates as coming from the scheduling, spectral, stop-time, and knapsack/transport papers.[10, 12, 13, 15] To keep the archive non-redundant, we also adopt the series contract-object interface (Synthesis 0): this paper exports an odds-inflation endpoint label, while other notes export schedule equalization, threat windows, and auditing objects that we consume via conversions.[1] We treat the bounded transcript projection and the repetition horizon as explicit claim fields: `exposure_nf_id` (ENF-ID) and `tw_id` (TW-ID), imported from the trace-interface and threat-window synthesis notes.[2, 3] Notation conventions for keys, state, transcripts, and transcript projections follow the series notation sheet (Synthesis 8).[25]

Exports. This paper exports (i) odds-inflation anonymity (Def. 1) as an endpoint metric, (ii) sequential/interactive composition bounds, and (iii) tight conversions from common equalization contracts (TV and moment bounds) into odds-inflation budgets. These outputs are designed to be “plugged into” Boss Fight accounting and certificate workflows elsewhere in the series.[4]

Artifact policy. This archive is source-only; supporting artifacts and tooling are available on request.[24]

1 Introduction

Anonymous systems (including anonymous DHT lookups) often come with a *stack* of mechanism-specific arguments: timing padding, cover traffic, routing randomization, batching, and other defenses. In our series, these are treated as modular “compiler” components for metadata-hiding overlays. This paper isolates a short, mechanism-agnostic endpoint: an upper bound on the *posterior odds* assigned to any identity after observing whatever transcript the system exposes.

Classical information-theoretic anonymity measures (e.g. entropy-based metrics) are valuable for intuition, but for certification we want a per-output bound that composes cleanly under interaction; odds inflation gives such an endpoint [5].

Non-redundancy with the series. We intentionally avoid re-deriving overlay- or routing-specific contracts here. Instead, we prove generic theorems showing how *any* component that exports a suitable privacy contract (likelihood-ratio caps, TV equalization, or moment bounds) can be converted into an odds-inflation guarantee that composes under interaction. Mechanism details (stop-time padding, scheduling compilers, spectral delegation bounds) are cited as prerequisites.

2 Model and definitions

Let $I \in \{1, \dots, N\}$ be a hidden identity with prior π and let O be any public observation (single output or full transcript). Write P_i for the law of O given $I = i$, and $P = \sum_i \pi(i)P_i$ for the mixture (the public marginal). We use $\text{TV}(P, Q) = \frac{1}{2}\|P - Q\|_1$ for total variation on finite alphabets.

Definition 1 (Per-identity and worst-case odds inflation). For any o with $P(o) > 0$, define the *per-identity odds inflation*

$$L_i(o) \stackrel{\text{def}}{=} \log_2 \frac{\pi(i | o)}{\pi(i)} = \log_2 \frac{P_i(o)}{P(o)}.$$

Define the *realized (worst-case) odds inflation*

$$\ell(o) \stackrel{\text{def}}{=} \max_{i \in \{1, \dots, N\}} L_i(o).$$

Definition 2 (ε_∞ and (ε, δ) odds-inflation anonymity). A mechanism is ε_∞ -*anonymous* (relative to prior π) if $\ell(o) \leq \varepsilon$ for all o with $P(o) > 0$. More generally, it is (ε, δ) -*anonymous* if for every event A ,

$$P_i(A) \leq 2^\varepsilon P(A) + \delta \quad \text{for all } i \in \{1, \dots, N\}. \tag{1}$$

Remark 1 (Interpretation). $\ell(o)$ is the number of *bits* by which the observation o can multiply the posterior odds of the most-favored identity (base 2). Results phrased in nats use $\ln$ and can be converted by dividing by $\ln 2$. The (ε, δ) form (1) is a tail envelope: with probability at most δ (under P_i), the realized loss can exceed ε by an arbitrary amount. Equivalently, (1) is the hockey-stick (E_γ) divergence bound $E_{2^\varepsilon}(P_i \| P) \leq \delta$, i.e., $\sup_A \{P_i(A) - 2^\varepsilon P(A)\} \leq \delta$ [21]. Hockey-stick divergence also underlies privacy-profile style accounting in differential privacy and remains an active object of study for auditing and restricted-adversary models (including “measured” hockey-stick divergence in quantum settings) [22]. This mirrors the usual (ε, δ) differential privacy inequality but with a

mixture baseline [19]. Related inferential-privacy frameworks (e.g., Pufferfish) are also phrased as posterior-odds ratio bounds between specified secret pairs; odds-inflation anonymity can be viewed as a one-vs-mixture specialization tailored to anonymity/attribution settings [20].

3 Priors and contexts as certificate inputs

Odds inflation is defined relative to a prior π because the public marginal $P = \sum_i \pi(i)P_i$ appears in the Bayes factor P_i/P . This is a feature: a deployment’s *context* (who could plausibly be the requester) is part of the privacy surface. Accordingly, in our series we treat priors and eligibility constraints as first-class certificate inputs (Anonymity C) [17].

Certification practice. A paper should state which prior or prior family its claimed budget targets. Two common, conservative patterns are:

1. **Uniform over an eligibility pool.** Declare an eligibility interface (e.g. “clients active in this epoch”), and take π uniform over that pool. Then the operational corollaries in Propositions 2 and 3 translate directly into lower bounds on the number of plausible requesters.
2. **Capped prior mass.** Declare a cap $\pi(i) \leq \pi_{\max}$ for all identities in scope. Under ε_∞ -anonymity, every posterior mass satisfies $\pi(i \mid o) \leq 2^\varepsilon \pi_{\max}$, so deployments can report a policy-friendly statement such as “no single identity exceeds p^* posterior mass” by enforcing $2^\varepsilon \pi_{\max} \leq p^*$.

In both cases, *eligibility metadata must be treated as a public interface* and budgeted (or hidden) rather than silently absorbed into the prior.

Proposition 1 (Eligibility/selection disclosures add a prior-shift term). *Let $C \subseteq \{1, \dots, N\}$ with $\pi(C) > 0$ and let $E \stackrel{\text{def}}{=} \mathbf{1}\{I \in C\}$ denote an eligibility/selection bit that becomes public (e.g. an observer learns which shard/committee the requester must belong to). Let $P_C(o) \stackrel{\text{def}}{=} \sum_{j \in C} \pi(j \mid C)P_j(o)$ be the public marginal conditioned on $I \in C$. Then for any $i \in C$ and any observation o with $P_C(o) > 0$, the realized odds inflation for the combined disclosure ($E=1, O=o$) satisfies*

$$\log_2 \frac{\pi(i \mid E=1, o)}{\pi(i)} = -\log_2 \pi(C) + \log_2 \frac{P_i(o)}{P_C(o)}.$$

In particular, if the mechanism is ε_∞ -anonymous relative to the conditional prior $\pi(\cdot \mid C)$ (i.e., $\max_{i \in C} \log_2 \frac{P_i(o)}{P_C(o)} \leq \varepsilon$ for all o), then relative to the original prior it is $(\varepsilon + \log_2 \frac{1}{\pi(C)})_\infty$ -anonymous for identities in C (equivalently, $\varepsilon - \log_2 \pi(C)$). Under a uniform prior this adds the familiar term $\log_2(N/|C|)$.

Proof. For $i \in C$, $P_i(E=1) = 1$ while $P(E=1) = \pi(C)$. Moreover $P(E=1, O=o) = \pi(C) P_C(o)$. Therefore $\log_2 \frac{P_i(E=1, O=o)}{P(E=1, O=o)} = \log_2 \frac{P_i(o)}{\pi(C)P_C(o)} = -\log_2 \pi(C) + \log_2 \frac{P_i(o)}{P_C(o)}$. Finally, $\pi(i \mid E=1, o)/\pi(i) = P_i(E=1, O=o)/P(E=1, O=o)$ by Bayes. $\square$

Remark 2 (Context-aware leakage bounds). Recent pointwise-maximal-leakage analyses show that even for fixed mechanisms, leakage can be substantially smaller under a committed prior model (e.g., minimum-probability assumptions) than under worst-case, context-free bounds [8]. Odds-inflation anonymity makes this dependence explicit by treating the prior (or prior family) as a certificate input.

4 Relation to PML/MaxL (pointer)

The realized odds inflation $\ell(o) = \max_i \log_2 \frac{P_i(o)}{P(o)}$ is exactly pointwise maximal leakage (PML) at outcome o under the same prior; maximal leakage (MaxL) is the log-moment of $\exp(\ell(O))$ and yields the standard $\log_2(\alpha K)$ sizing rule for exact identification. To keep this paper non-redundant, we treat these equivalences as “owned” by the endpoint-bridge synthesis note and the Boss-Fight budgets paper; see [4, 26].

5 Closure properties

5.1 Post-processing

Theorem 1 (Post-processing). *Let $Z = g(O)$ be any (possibly randomized) function of O . Then for every realized value z ,*

$$\ell_Z(z) \leq \operatorname{ess\,sup}_{o: K(z|o) > 0} \ell_O(o),$$

where K is the Markov kernel of g (for deterministic g , this is $\sup_{o: g(o)=z} \ell_O(o)$). In particular, ε_∞ -anonymity and (ε, δ) -anonymity are preserved under post-processing.

Proof. Fix z with $P_Z(z) > 0$. For each i , write the Bayes factor $B_i(o) \stackrel{\text{def}}{=} P_i(o)/P(o)$. Then

$$\frac{P_i(Z = z)}{P(Z = z)} = \frac{\sum_o P(o) B_i(o) K(z | o)}{\sum_o P(o) K(z | o)} = \mathbb{E}_P[B_i(O) | Z = z],$$

where K is the Markov kernel of g (randomized post-processing) and the conditional expectation is under the mixture P . Therefore

$$\max_i \frac{P_i(Z = z)}{P(Z = z)} \leq \mathbb{E}_P \left[\max_i B_i(O) \mid Z = z \right].$$

Since a conditional expectation is bounded by the essential supremum on the conditioning event,

$$\mathbb{E}_P \left[\max_i B_i(O) \mid Z = z \right] \leq \operatorname{ess\,sup}_{o: K(z|o) > 0} \max_i B_i(o) = \exp \left(\operatorname{ess\,sup}_{o: K(z|o) > 0} \ell_O(o) \right).$$

Taking logs yields the stated bound. For (ε, δ) , note that (1) is the E_{2^ε} (hockey-stick) divergence bound $E_{2^\varepsilon}(P_i \| P) \leq \delta$, and E_γ satisfies data processing under Markov kernels, so the same (ε, δ) pair holds for Z [21]. $\square$

5.2 Sequential composition under interaction

Consider an interactive transcript $O_{1:T}$ where each step may depend on the past. Write ℓ_t for the realized odds inflation after observing prefix $O_{1:t}$.

Theorem 2 (Sequential composition for ε_∞). *If each step t satisfies the conditional per-step bound*

$$\max_i \log_2 \frac{P_i(O_t | O_{1:t-1})}{P(O_t | O_{1:t-1})} \leq \varepsilon_t \quad \text{almost surely,}$$

then the full transcript is $(\sum_{t=1}^T \varepsilon_t)_\infty$ -anonymous.

Proof. Let $B_{i,t} \stackrel{\text{def}}{=} \frac{P_i(O_{1:t})}{P(O_{1:t})}$ be the Bayes factor for identity i after t steps. By the chain rule,

$$B_{i,T} = \prod_{t=1}^T \frac{P_i(O_t | O_{1:t-1})}{P(O_t | O_{1:t-1})}.$$

Taking $\log_2$, maximizing over i , and applying the per-step bound yields $\ell(O_{1:T}) = \log_2 \max_i B_{i,T} \leq \sum_{t=1}^T \varepsilon_t$ almost surely. $\square$

Theorem 3 (Sequential composition for (ε, δ) checkpoints). *Consider an interactive transcript $O_{1:T}$. Assume that for each step t there exist numbers $(\varepsilon_t, \delta_t)$ such that, almost surely over the public history $O_{1:t-1}$, for every event A and every identity i ,*

$$P_i(O_t \in A | O_{1:t-1}) \leq 2^{\varepsilon_t} P(O_t \in A | O_{1:t-1}) + \delta_t, \quad (2)$$

where $P(\cdot | O_{1:t-1})$ is the mixture (posterior) baseline at step t . Then the full transcript satisfies, for every event B and every identity i ,

$$P_i(O_{1:T} \in B) \leq 2^\varepsilon P(O_{1:T} \in B) + \delta_{\text{tot}}, \quad \varepsilon := \sum_{t=1}^T \varepsilon_t, \quad \delta_{\text{tot}} := \sum_{t=1}^T \delta_t \exp\left(\sum_{s < t} \varepsilon_s\right).$$

In particular, if $\varepsilon_t \geq 0$ for all t , then $\delta_{\text{tot}} \leq 2^\varepsilon \sum_t \delta_t$.

Proof. We prove the bound by induction on T . For $T = 1$ it is (2). Assume it holds up to $T - 1$. Fix an event B on $O_{1:T}$ and write its sections $B_h := \{o_T : (h, o_T) \in B\}$ for histories $h = o_{1:T-1}$. By the tower property,

$$P_i(B) = \mathbb{E}_{P_i}[P_i(O_T \in B_{O_{1:T-1}} | O_{1:T-1})].$$

Applying (2) with $A = B_{O_{1:T-1}}$ yields

$$P_i(B) \leq 2^{\varepsilon_T} \mathbb{E}_{P_i}[\phi(O_{1:T-1})] + \delta_T, \quad \phi(h) := P(O_T \in B_h | O_{1:T-1} = h).$$

To relate $\mathbb{E}_{P_i}[\phi(O_{1:T-1})]$ to $\mathbb{E}_P[\phi(O_{1:T-1})] = P(B)$, use the layer-cake identity $\phi = \int_0^1 \mathbf{1}\{\phi \geq u\} du$ to reduce the claim to events and apply the induction hypothesis to each threshold set $\{\phi \geq u\}$. This yields

$$\mathbb{E}_{P_i}[\phi(O_{1:T-1})] \leq 2^{\sum_{t < T} \varepsilon_t} \mathbb{E}_P[\phi(O_{1:T-1})] + \sum_{t < T} \delta_t \exp\left(\sum_{s < t} \varepsilon_s\right).$$

Combining and simplifying yields the stated δ_{tot} . $\square$

Corollary 1 (Homogeneous checkpoints). *If the per-step bound (2) holds with the same pair $(\varepsilon_t, \delta_t) \equiv (\varepsilon, \delta)$ for all t (uniformly over histories), then for every event B and every identity i ,*

$$P_i(O_{1:T} \in B) \leq 2^{T\varepsilon} P(O_{1:T} \in B) + \delta \sum_{j=0}^{T-1} 2^{j\varepsilon} = 2^{T\varepsilon} P(B) + \begin{cases} \delta \frac{2^{T\varepsilon} - 1}{2^\varepsilon - 1}, & \varepsilon \neq 0, \\ T\delta, & \varepsilon = 0. \end{cases}$$

In particular, if $\varepsilon \geq 0$ then the slack is at most $2^{T\varepsilon} T\delta$.

Remark 3 (Anytime-valid monitoring). The per-identity Bayes factor process $B_{i,t} = \exp(L_i(O_{1:t}))$ is a nonnegative martingale under the mixture law. This connects odds inflation to e -processes and time-uniform bounds (Ville-style inequalities), and to privacy odometers/filters in adaptive DP composition. See [7, 6].

6 Operational consequences

Odds caps imply concrete limits on what any adversary can do with the observation.

Lemma 1 (Suspect sets must have prior mass). *For any threshold $\tau > 0$ and any observation o ,*

$$\pi(\{i : L_i(o) \geq \log_2 \tau\}) \leq \frac{1}{\tau}.$$

In particular, if $\ell(o) \leq \varepsilon$ then every identity i satisfies $\pi(i | o) \leq 2^\varepsilon \pi(i)$.

Proposition 2 (Shortlists and credible sets). *Suppose the mechanism is ε_∞ -anonymous. Let $S(o) \subseteq \{1, \dots, N\}$ be any post-processed shortlist output on observation o . If $S(o)$ is required to have posterior coverage $\pi(S(o) | o) \geq 1 - \beta$ for all o , then $\pi(S(o)) \geq 2^{-\varepsilon}(1 - \beta)$ for all o . Under a uniform prior this forces $|S(o)| \geq (1 - \beta)2^{-\varepsilon}N$.*

Proof. Fix any observation o with $P(o) > 0$. By ε_∞ -anonymity, for every identity i we have $\pi(i | o) \leq 2^\varepsilon \pi(i)$. Summing over $i \in S(o)$ gives

$$\pi(S(o) | o) = \sum_{i \in S(o)} \pi(i | o) \leq 2^\varepsilon \sum_{i \in S(o)} \pi(i) = 2^\varepsilon \pi(S(o)).$$

If $\pi(S(o) | o) \geq 1 - \beta$, then rearranging yields $\pi(S(o)) \geq 2^{-\varepsilon}(1 - \beta)$. For the uniform prior, $\pi(S(o)) = |S(o)|/N$, hence $|S(o)| \geq (1 - \beta)2^{-\varepsilon}N$. $\square$

Corollary 2 (Exact identification is bounded). *Assume a uniform prior on $\{1, \dots, N\}$ and ε_∞ -anonymity. Then for any estimator $\hat{I} = \hat{I}(O)$,*

$$\mathbb{P}(\hat{I} = I) \leq \frac{2^\varepsilon}{N}.$$

Proof. Condition on $O = o$. The best success probability is the MAP mass $\max_i \pi(i | o)$, which is at most $2^\varepsilon/N$ under ε_∞ -anonymity and the uniform prior. Average over o . $\square$

Corollary 3 (Identification under (ε, δ)). *Assume a uniform prior on $\{1, \dots, N\}$ and (ε, δ) -anonymity in the sense of Definition 2. Then for any estimator $\hat{I} = \hat{I}(O)$,*

$$\mathbb{P}(\hat{I} = I) \leq \frac{2^\varepsilon}{N} + \delta.$$

Proof. Let $A_i = \{o : \hat{I}(o) = i\}$. Under the uniform prior, $\mathbb{P}(\hat{I} = I) = \frac{1}{N} \sum_{i=1}^N P_i(A_i)$. Apply (1) to each A_i to obtain $P_i(A_i) \leq 2^\varepsilon P(A_i) + \delta$. Averaging over i and using $\sum_i P(A_i) = 1$ yields the claim. $\square$

Proposition 3 (Property inference is also capped). *Assume ε_∞ -anonymity. For any subset $E \subseteq \{1, \dots, N\}$ and any observation o with $P(o) > 0$,*

$$\pi(E | o) \leq 2^\varepsilon \pi(E).$$

Equivalently, no post-processed binary property of the identity can have its posterior probability inflated by more than 2^ε .

Proof. Sum the pointwise bounds $\pi(i | o) \leq 2^\varepsilon \pi(i)$ over $i \in E$. $\square$

7 Practical conversions from common contracts

This section records two conversion lemmas that let mechanism-specific analyses “plug into” odds inflation. The TV-equalization form is the same interface used by our TV-privacy padding compilers and dual-certificate tooling [15].

7.1 TV equalization to a smoothed reference

Let Q be a public reference distribution on outcomes with $Q_{\min} := \min_o Q(o) > 0$. In many anonymity channels, a mechanism exports a *common-reference* guarantee: both the per-identity output law P_i and the public marginal $P = \sum_i \pi(i)P_i$ are close to the same Q (e.g. via equalization, smoothing, or explicit cover traffic). This suffices to bound the identity-vs-mixture Bayes factors P_i/P that define odds inflation.

Lemma 2 (Common-reference TV-to-likelihood-ratio). *Let P and R be distributions on a finite alphabet, and let Q satisfy $Q_{\min} > 0$. If $\text{TV}(P, Q) \leq \delta$ and $\text{TV}(R, Q) \leq \delta$ with $\delta < Q_{\min}/2$, then for all outcomes o ,*

$$\left| \log_2 \frac{P(o)}{R(o)} \right| \leq \log_2 \frac{Q_{\min} + 2\delta}{Q_{\min} - 2\delta}.$$

Proof. Since $\|P - Q\|_1 \leq 2\delta$, for each o we have $P(o) \leq Q(o) + 2\delta$. Similarly $\|R - Q\|_1 \leq 2\delta$ implies $R(o) \geq Q(o) - 2\delta$. Therefore

$$\frac{P(o)}{R(o)} \leq \frac{Q(o) + 2\delta}{Q(o) - 2\delta} \leq \frac{Q_{\min} + 2\delta}{Q_{\min} - 2\delta},$$

where the last step uses that $(x + 2\delta)/(x - 2\delta)$ is decreasing for $x > 2\delta$. The reverse ratio $R(o)/P(o)$ is handled symmetrically. $\square$

Corollary 4 (Equalization $\Rightarrow$ per-output odds-inflation cap). *If a mechanism satisfies $\text{TV}(P_i, Q) \leq \delta$ for all identities i (with the same Q and δ), then $\text{TV}(P, Q) \leq \delta$ for the public marginal $P = \sum_i \pi(i)P_i$. Consequently, if $\delta < Q_{\min}/2$ then for every realized outcome o ,*

$$\ell(o) = \max_i \log_2 \frac{P_i(o)}{P(o)} \leq \log_2 \frac{Q_{\min} + 2\delta}{Q_{\min} - 2\delta}.$$

Proof. By convexity of TV, $\text{TV}(P, Q) = \text{TV}(\sum_i \pi(i)P_i, Q) \leq \sum_i \pi(i) \text{TV}(P_i, Q) \leq \delta$. Apply Lemma 2 with $P = P_i$ and $R = P$ and maximize over i . $\square$

Remark 4. This conversion is conservative but operationally important: without a positive background mass ($Q_{\min} = 0$), worst-case odds inflation can be infinite. For anonymous DHTs, this is the “rare committee” failure mode. When $Q_{\min}$ is too small to enforce, use a trimmed-support tail guarantee instead (next lemma).

Lemma 3 (Common-reference trimmed-support envelope). *Let $\tau > 0$ and define the “well-supported” set $S_\tau \stackrel{\text{def}}{=} \{o : Q(o) \geq \tau\}$. If $\text{TV}(P, Q) \leq \delta$ and $\text{TV}(R, Q) \leq \delta$ with $\delta < \tau/2$, then for all $o \in S_\tau$,*

$$\left| \log_2 \frac{P(o)}{R(o)} \right| \leq \log_2 \frac{\tau + 2\delta}{\tau - 2\delta}.$$

Moreover, $P(S_\tau^c) \leq Q(S_\tau^c) + \delta$ and $R(S_\tau^c) \leq Q(S_\tau^c) + \delta$. Thus a TV equalization contract yields a finite loss bound on S_τ and pushes the remaining probability mass into explicit “rare outcome” tail terms.

7.2 Moment (Rényi) bounds to tail envelopes

Mechanism analyses often bound conditional Rényi divergences or log-moments of the privacy loss. Such bounds convert directly into (ε, δ) odds-inflation envelopes by a one-line Chernoff argument.

Lemma 4 (Rényi-to-tail conversion). *Fix an identity i and let $L_i(o) = \log_2 \frac{P_i(o)}{P(o)}$ be the per-output privacy loss. If for some order $\alpha > 1$,*

$$D_\alpha(P_i \| P) \leq \rho_\alpha,$$

then for every threshold ε ,

$$P_i(L_i(O) \geq \varepsilon) \leq \exp((\alpha - 1)(\rho_\alpha - \varepsilon)).$$

Equivalently, with $\delta = \exp((\alpha - 1)(\rho_\alpha - \varepsilon))$, the odds inflation exceeds ε with probability at most δ under P_i .

Proof. By definition of Rényi divergence and the identity $\exp((\alpha - 1)L_i) = (P_i/P)^{\alpha-1}$,

$$\mathbb{E}_{P_i}[\exp((\alpha - 1)L_i(O))] = \sum_o P_i(o) \left(\frac{P_i(o)}{P(o)} \right)^{\alpha-1} = \sum_o P_i(o)^\alpha P(o)^{1-\alpha} = \exp((\alpha-1)D_\alpha(P_i \| P)) \leq 2^{(\alpha-1)\rho_\alpha}.$$

Applying Markov’s inequality under P_i gives

$$P_i(L_i \geq \varepsilon) = P_i(\exp((\alpha - 1)L_i) \geq 2^{(\alpha-1)\varepsilon}) \leq 2^{(\alpha-1)(\rho_\alpha - \varepsilon)}.$$

□

Remark 5 (Connection to RDP/moment accountants). Lemma 4 is the same tail conversion that underlies Rényi differential privacy (RDP) and moment-accountant style envelopes, but applied to the identity-vs-mixture loss L_i rather than neighboring datasets. See [18] for the RDP framing, [19] for differential privacy background, and [6] for adaptive composition tooling.

8 A worked audit-style example (telemetry to odds inflation)

Suppose a deployment fixes a discrete interface (output alphabet of size K) and a public background Q with floor $Q_{\min}$ (or a trimmed floor τ). For each audited identity/class i (or each equivalence class of identities), an auditor collects a window of m i.i.d. samples from the interface and forms an empirical distribution $\hat{P}_i$. A conservative L_1 deviation bound of Weissman et al. yields a confidence radius for $\|\hat{P}_i - P_i\|_1$ from finite samples [23]. Translating to TV gives an upper confidence bound on $\text{TV}(P_i, Q)$ when Q is known. Once all audited P_i satisfy $\text{TV}(P_i, Q) \leq \delta$, Corollary 4 converts this equalization certificate into an odds-inflation cap against the public marginal.

Beyond TV radii (optional). If a deployment can support black-box hypothesis testing between P_i and the mixture P , one can audit richer tradeoff curves (e.g., f -DP / tradeoff functions) rather than a single TV radius; see Askin et al. for recent black-box f -DP estimation and auditing methods [9].

For a fully worked, receipt-style instantiation (including ABOM/OINL and line-item semantics), see Synthesis 17.[27]

This “telemetry $\rightarrow$ equalization $\rightarrow$ odds inflation” pipeline is what Anonymity B specializes to anonymous DHT committee traces, and what Anonymity C records in an ABOM certificate [16, 17].

9 How to consume this paper in the series

Once a module exports either (i) a likelihood-ratio cap, (ii) TV equalization to a smoothed reference, or (iii) a moment bound, the odds-inflation endpoint and its composition rules apply verbatim. This allows each mechanism paper to avoid re-proving downstream operational consequences (shortlisting limits, identification bounds, etc.) and simply cite this paper. For timing/termination-time contracts in anonymous overlay lookups, see the companion mechanism toolboxes [10, 11]; for delegation-style overlays, see [12].

10 Takeaways

- **Endpoint metric:** odds inflation is a posterior-odds cap for a transcript projection X and a declared prior/context. It is the right “final number” to publish when the operational question is *how well can an adversary decide?*
- **Composable by default:** sequential use over epochs composes additively in log-odds (and via standard odometer/filter reasoning when stopping rules are adaptive). Mechanism papers in this archive should export per-epoch budgets and cite this paper for the long-horizon bound.
- **Operational conversions:** odds budgets induce (i) shortlist limits and (ii) identification sizing rules of the form $\log_2(\alpha K)$, enabling “privacy-to-knobs” compilation without re-deriving decision-theoretic consequences (see also Synthesis 5).[4]
- **Certificate interface:** the certificate inputs are the transcript projection, adjacency notion, and budget representation; ABOM/OINL (Anonymity C) is the canonical carrier for committing to these semantics in a source-only release.[17]

References

- [1] Anonymous. Contract Objects for Anonymous-DHT Privacy Budgets and Claims. Series synthesis note (Synthesis 0), 2026. (This archive.)
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